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Outboard motor and anti-vibration structure of outboard motor

Abstract

An outboard motor includes an outboard motor main body and a mount assembly to mount the outboard motor main body to a hull. The mount assembly includes a clamp bracket to be fixed to the hull, a swivel bracket to support the outboard motor main body, and a hydraulic cylinder. The swivel bracket is coupled to the clamp bracket through a tilt shaft and turnable around the tilt shaft. The hydraulic cylinder is located between the clamp bracket and the swivel bracket and is able to turn the swivel bracket around the tilt shaft with respect to the clamp bracket. The outboard motor includes a gas damper to damp an external force that acts in a telescopic direction of the hydraulic cylinder.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2021-185232 filed on Nov. 12, 2021. The entire contents of this application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to an outboard motor and an anti-vibration structure of an outboard motor.

2. Description of the Related Art

(3) Japanese Laid-open Patent Application Publication No. H9-11988 discloses a tilt/trim device of a vessel propulsion apparatus. This device is configured so that hydraulic oil is supplied from a pumping device to a hydraulic cylinder device located between a swivel bracket of an outboard

motor and a hull. A propulsion unit of the outboard motor is tilt-operated or trim-operated by the extension and contraction of the hydraulic cylinder device.

SUMMARY OF THE INVENTION

(4) The inventor of preferred embodiments of the present invention described and claimed in the present application conducted an extensive study and research regarding an outboard motor, such as the one described above, and in doing so, discovered and first recognized new unique challenges and previously unrecognized possibilities for improvements as described in greater detail below.

(5) An engine rotates at a low speed during low-speed navigation, such as trolling. At this time, low-frequency vibrations are transmitted from the outboard motor to a hull. If these vibrations can be reduced, a comfortable ride on the vessel can be improved.

(6) Preferred embodiments of the present invention provide outboard motors that each significantly reduces or prevents vibrations transmitted to a hull, and anti-vibration structures of the outboard motors.

(7) In order to overcome the previously unrecognized and unsolved challenges described above, a preferred embodiment of the present invention provides an outboard motor including an outboard motor main body and a mount assembly to mount the outboard motor main body to a hull. The mount assembly includes a clamp bracket to be fixed to the hull and a swivel bracket that is coupled to the clamp bracket through a tilt shaft, turnable around the tilt shaft, and supports the outboard motor main body. The outboard motor further includes a hydraulic cylinder located between the clamp bracket and the swivel bracket and that is able to turn the swivel bracket around the tilt shaft with respect to the clamp bracket. The outboard motor further includes a gas damper to damp an external force that acts in a telescopic direction of the hydraulic cylinder. According to this structural configuration, an external force that acts in the telescopic direction of the hydraulic cylinder that turns the swivel bracket around the tilt shaft is attenuated by the gas damper. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(8) In a preferred embodiment of the present invention, the hydraulic cylinder includes a cylinder main body, a piston that is slidable in the cylinder main body and that defines a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers. The hydraulic cylinder extends and contracts the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers. Additionally, the gas damper is provided in the cylinder main body and is filled with gas. According to this structural configuration, the gas damper functions as a damper, and, as a result, an external force that acts in the telescopic direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(9) In a preferred embodiment of the present invention, the gas damper includes a free piston that is slidable in the cylinder main body on a side opposite to the rod and that defines a gas chamber filled with the gas in the cylinder main body. According to this structural configuration, gas with which the gas chamber defined by the free piston in the cylinder main body of the hydraulic cylinder is filled functions as a damper, and, as a result, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(10) In a preferred embodiment of the present invention, the cylinder main body includes a contact portion that is contactable with the free piston so as to regulate a position of a stroke end of the free piston and that regulates a minimum volume of the gas chamber. According to this structural configuration, an increase of the pressure in the gas chamber is regulated.

(11) In a preferred embodiment of the present invention, the hydraulic cylinder includes a cylinder main body, a piston that is slidable in the cylinder main body and that defines a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers. The hydraulic cylinder extends and contracts the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers. The gas

damper includes a sub-cylinder that is hydraulically connected to at least one of the pair of oil chambers.

(12) According to this structural configuration, the sub-cylinder that is hydraulically connected to at least one of the oil chambers of the hydraulic cylinder functions as a gas damper, and, as a result, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(13) In a preferred embodiment of the present invention, the sub-cylinder includes a sub-cylinder main body and a free piston that is slidable in the sub-cylinder main body. The free piston divides a sub-oil chamber in communication with at least one of the pair of oil chambers of the hydraulic cylinder and a gas chamber filled with the gas from each other. According to this structural configuration, the gas with which the gas chamber divided from the sub-oil chamber by the free piston in the sub-cylinder main body is filled functions as a damper, and, as a result, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(14) In a preferred embodiment of the present invention, the sub-cylinder main body includes a contact portion that is contactable with the free piston so as to regulate a position of a stroke end of the free piston and that regulates a minimum volume of the gas chamber. According to this structural configuration, an increase of the pressure in the gas chamber is regulated.

(15) In a preferred embodiment of the present invention, the sub-cylinder includes a pair of sub-cylinders. Sub-oil chambers of the pair of sub-cylinders are connected to the pair of oil chambers of the hydraulic cylinder, respectively. According to this structural configuration, each of the sub-cylinders that respectively correspond to the extension and the contraction of the hydraulic cylinder functions as a gas damper. Thus, it is possible to attenuate an extension/contraction bidirectional external force. Therefore, the attenuation effect is high. This makes it possible to further significantly reduce or prevent vibrations transmitted to the hull.

(16) In a preferred embodiment of the present invention, the sub-cylinder includes only a single sub-cylinder, and both of the pair of oil chambers of the hydraulic cylinder are in communication with the sub-oil chamber of the single sub-cylinder. According to this structural configuration, the sub-cylinder common to the extension and the contraction of the hydraulic cylinder is able to function as a gas damper.

(17) In a preferred embodiment of the present invention, the sub-cylinder includes only a single sub-cylinder, and the gas chamber includes a pair of gas chambers, and the sub-oil chamber includes a pair of sub-oil chambers. The pair of oil chambers of the hydraulic cylinder are in communication with the sub-oil chambers corresponding to the pair of oil chambers, respectively. The free piston includes a rod piston that includes a gas piston that divides the pair of gas chambers from each other and a pair of rods that extend from the gas piston to both sides in a sliding direction of the free piston. Each of the pair of rods functions as a hydraulic piston that defines a portion of the sub-oil chamber corresponding to the rod. According to this structural configuration, the sub-cylinder common to the extension and the contraction of the hydraulic cylinder is able to function as a gas damper.

(18) In a preferred embodiment of the present invention, the outboard motor further includes an on-off valve to interrupt a flow of hydraulic oil between at least one of the pair of oil chambers of the hydraulic cylinder and the sub-oil chamber of the sub-cylinder. According to this structural configuration, gas with which the gas chamber of the sub-cylinder is filled functions as a damper in a state in which a flow of hydraulic oil between at least one of the pair of oil chambers of the hydraulic cylinder and the sub-oil chamber of the sub-cylinder is allowed by the on-off valve. Thus, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

- (19) In a preferred embodiment of the present invention, the outboard motor further includes a prime mover to rotate a propeller, a detector to detect a rotation speed of the prime mover, and a controller configured or programmed to shut off a control valve, which is the on-off valve, when a rotation speed detected by the detector exceeds a predetermined threshold value. According to this structural configuration, gas with which the gas chamber of the sub-cylinder is filled functions as a damper when a detection value of the rotation speed of the prime mover is lower than the threshold value. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.
- (20) In a preferred embodiment of the present invention, the hydraulic cylinder includes a cylinder main body, a piston that is slidable in the cylinder main body and that defines a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers. The hydraulic cylinder extends and contracts the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers. Additionally, the gas damper includes a gas cylinder that slidably houses an end portion of the cylinder main body on a side opposite to the rod and that defines a gas chamber, filled with gas, with the end portion of the cylinder main body.
- (21) According to this structural configuration, the gas cylinder that slidably houses the end portion of the cylinder main body on the side opposite to the rod functions as a gas damper, and, as a result, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.
- (22) In a preferred embodiment of the present invention, the gas cylinder includes a contact portion that is contactable with the end portion of the cylinder main body. The contact portion regulates a minimum volume of the gas chamber by regulating a position of a stroke end of the end portion of the cylinder main body with respect to the gas cylinder. According to this structural configuration, an increase of the pressure in the gas chamber is regulated.
- (23) In a preferred embodiment of the present invention, the hydraulic cylinder includes a cylinder main body, a piston that is slidable in the cylinder main body and that defines a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers. The hydraulic cylinder extends and contracts the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers. Additionally, the piston includes a hollow piston having a cylindrical inner peripheral surface that defines a gas chamber filled with gas. Additionally, the gas damper includes a double piston structure including the hollow piston and an internal piston that is housed in the hollow piston slidably on the inner peripheral surface, coupled to the rod, and partitions the gas chamber into a pair of gas chambers.
- (24) According to this structural configuration, the internal piston performs a stroke in the hollow piston in response to the extension/contraction of the rod, and, as a result, gas in a corresponding one of the gas chambers in the hollow piston is compressed. Thus, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.
- (25) In a preferred embodiment of the present invention, the hollow piston includes a contact portion that is contactable with the internal piston and that regulates a minimum volume of the gas chamber by regulating a position of a stroke end of the internal piston with respect to the hollow piston. According to this structural configuration, an increase of the pressure in the gas chamber is regulated.
- (26) In a preferred embodiment of the present invention, the hydraulic cylinder includes a cylinder main body, a piston that is slidable in the cylinder main body and that defines a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers. The hydraulic cylinder extends and contracts the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers.

Additionally, the gas damper includes a pair of independent foam structures located in the pair of oil chambers, respectively. According to this structural configuration, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated by the independent foam structure located in each of the oil chambers. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(27) In a preferred embodiment of the present invention, the hydraulic cylinder includes a cylinder main body, a piston that is slidable in the cylinder main body and that defines a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers. The hydraulic cylinder extends and contracts the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers.

Additionally, the gas damper includes a movable partition that defines a gas chamber filled with gas in the hydraulic cylinder or in a sub-cylinder that is hydraulically connected to the hydraulic cylinder.

(28) According to this structural configuration, gas, for example, in the gas chamber defined by the movable partition in the hydraulic cylinder functions as a damper, and, as a result, an external force that acts in the extension/contraction direction of the hydraulic cylinder is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(29) In a preferred embodiment of the present invention, the mount assembly further includes an elastic mount that elastically supports the outboard motor main body with respect to the swivel bracket, and the gas damper functions as a gas spring having a spring constant lower than a spring constant of the elastic mount. According to this structural configuration, it becomes possible to extend and contract the gas damper even when an external force that acts in the extension/contraction direction of the hydraulic cylinder is small. Therefore, it is possible to improve an anti-vibration effect, and it is possible to further significantly reduce or prevent vibrations transmitted to the hull.

(30) In another preferred embodiment of the present invention, an anti-vibration structure of an outboard motor includes a hydraulic cylinder that is able to turn an outboard motor main body around a tilt shaft coupled to a clamp bracket fixed to a hull and a gas damper to damp an external force that acts in a telescopic direction of the hydraulic cylinder. According to this structural configuration, an external force acting in the extension/contraction direction of the hydraulic cylinder that turns the outboard motor main body is attenuated by the gas damper. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(31) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side view shown to describe an example of an outboard motor according to a preferred embodiment of the present invention.

(2) FIG. 2 is a descriptive view showing a state in which an outboard motor main body has been tilted up.

(3) FIG. 3 is a schematic view of a mount assembly that attaches the outboard motor main body to a vessel.

(4) FIG. 4 is a cross-sectional view showing a cross section along line IV-IV of FIG. 1.

(5) FIG. 5A is a schematic view shown to describe an example of an anti-vibration structure of the outboard motor.

(6) FIG. 5B is a schematic cross-sectional view that describes a function in the example of the anti-

vibration structure of the outboard motor of FIG. 5A.

(7) FIG. 6A is a schematic view shown to describe another example of an anti-vibration structure of the outboard motor.

(8) FIG. 6B is a schematic cross-sectional view that describes a function in the example of the anti-vibration structure of the outboard motor of FIG. 6A.

(9) FIG. 7 is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(10) FIG. 8 is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(11) FIG. 9 is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(12) FIG. 10 is a schematic view shown to describe still another example of the anti-vibration structure of the outboard motor.

(13) FIG. 11 is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(14) FIG. 12 is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(15) FIG. 13 is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(16) FIG. 14A is a schematic view shown to describe still another example of an anti-vibration structure of the outboard motor.

(17) FIG. 14B is a schematic cross-sectional view that describes a function in the example of the anti-vibration structure of the outboard motor of FIG. 14A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) FIG. 1 is a side view shown to describe an example of an outboard motor 1 according to a preferred embodiment of the present invention. The outboard motor 1 includes an outboard motor main body 2, a mount assembly 3, and a power trim & tilt mechanism 4 (hereinafter, referred to as "PTT mechanism 4"). The outboard motor main body 2 is attached to a rear portion of a hull 5a of a vessel 5 by the mount assembly 3.

(19) The mount assembly 3 includes a swivel bracket 7, a clamp bracket 6, a steering shaft 8, a tilt shaft 9, an upper bracket 10, a lower bracket 11, and a plurality of anti-vibration mounts 12. The steering shaft 8 extends in an up-down direction. The tilt shaft 9 extends horizontally or substantially horizontally in a left-right direction (direction perpendicular to the plane of paper of FIG. 1). The swivel bracket 7 is coupled to the outboard motor main body 2 through the steering shaft 8. The PTT mechanism 4 is an example of a turning mechanism that turns the outboard motor main body 2 around the tilt shaft 9 and tilts the outboard motor main body 2 with respect to the clamp bracket 6. The anti-vibration mounts 12 are located on the upper and lower sides. The anti-vibration mount 12 is an example of an elastic mount that elastically supports the outboard motor main body 2 with respect to the swivel bracket 7.

(20) The upper bracket 10 and the lower bracket 11 are each an example of a bracket by which the outboard motor main body 2 is attached to the hull 5a. The upper bracket 10 is fixed to an upper end portion 8a of the steering shaft 8. The lower bracket 11 is fixed to a lower end portion 8b of the steering shaft 8. The upper bracket 10 and the lower bracket 11 are each coupled to the outboard motor main body 2 through the anti-vibration mount 12.

(21) The outboard motor main body 2 is attached to the hull 5a in a substantially vertical attitude by the mount assembly 3. The clamp bracket 6 is fixed to the hull 5a. The swivel bracket 7 is coupled to the clamp bracket 6 through the tilt shaft 9 so as to turn around the tilt shaft 9, and supports the outboard motor main body 2. The outboard motor main body 2 and the swivel bracket 7 are turnable around the tilt shaft 9 upwardly and downwardly with respect to the clamp bracket 6. The outboard motor main body 2 and the swivel bracket 7 are turned around the tilt shaft 9

upwardly and downwardly by the PTT mechanism **4**. The PTT mechanism **4** is actuated by operating a PTT operation switch (not shown). Therefore, the outboard motor main body **2** is tilted with respect to the clamp bracket **6** by the operation of the PTT operation switch. As a result, it is possible to change the inclination angle of the outboard motor main body **2** with respect to the hull **5a**, and therefore it is possible to make a trim adjustment and to tilt up/down the outboard motor main body **2**. The outboard motor **1** includes an anti-vibration structure BS of the outboard motor (hereinafter, referred to simply as an “anti-vibration structure BS” if necessary) that fulfills a function to significantly reduce or prevent vibrations transmitted to the hull **5a** by use of a portion of the PTT mechanism **4**.

(22) The PTT mechanism **4** includes a trim cylinder **30** to make a trim adjustment and a tilt cylinder **40** to tilt up/down the outboard motor main body **2**. The trim cylinder **30** includes a cylinder main body **31** and a trim rod **32**. The tilt cylinder **40** includes a cylinder main body **41** and a tilt rod **42**. The tilt cylinder **40** is joined between the clamp bracket **6** and the swivel bracket **7**. The tilt cylinder **40** turns the swivel bracket **7** around the tilt shaft **9** with respect to the clamp bracket **6**. The tilt cylinder **40** is an example of a hydraulic cylinder HC applied to the anti-vibration structure BS. The anti-vibration structure BS includes a gas damper GD to damp an external force that acts in an expansion/contraction direction X (a telescopic direction) of the hydraulic cylinder HC.

(23) Additionally, the outboard motor main body **2** is turnable around the steering shaft **8** rightwardly and leftwardly with respect to the swivel bracket **7**. A steering wheel (not shown) is operated, and, as a result, the outboard motor main body **2** is turned around the steering shaft **8** rightwardly and leftwardly. This makes it possible to steer the vessel **5**.

(24) The outboard motor main body **2** includes an engine **13** as an example of a prime mover, a drive shaft **14**, a propeller shaft **15**, a propeller **16** as an example of a thrust generating member, a forward-reverse switching mechanism **17** as an example of a clutch, and an ECU (Electronic Control Unit) **18** as an example of a controller. Additionally, the outboard motor main body **2** includes an engine cover **19** and a casing **20**. The engine **13** and the ECU **18** are housed in the engine cover **19**. Additionally, the drive shaft **14** extends upwardly and downwardly in the engine cover **19** and the casing **20**. The propeller shaft **15** extends in a front-rear direction in a lower portion of the casing **20**. An upper end portion of the drive shaft **14** is coupled to the engine **13**. A lower end portion of the drive shaft **14** is coupled to a front end portion of the propeller shaft **15** by the forward-reverse switching mechanism **17**. The propeller **16** is coupled to a rear end portion of the propeller shaft **15**. The propeller **16** rotates together with the propeller shaft **15**. The propeller **16** is rotationally driven by the engine **13**. In other words, the engine **13** rotates the propeller **16**.

(25) The engine **13** may be, for example, an internal combustion engine that burns fuel, such as gasoline, to generate power. The engine **13** includes a crankshaft **21**, a plurality of (for example, four) cylinders **22**, and a rotation speed detector **23**. The engine **13** is oriented such that the crankshaft **21** extends vertically. The upper end portion of the drive shaft **14** is joined to the crankshaft **21**. The crankshaft **21** is rotationally driven around a vertical axis by combustion in each of the cylinders **22**. The rotation speed of the crankshaft **21** (rotation speed of the engine **13**) is detected by both the rotation speed detector **23** and the ECU **18**. The rotation speed detector **23** outputs a detection signal that synchronizes with the rotation of the crankshaft **21**. The ECU **18** calculates the engine rotation speed based on the detection signal.

(26) FIG. 2 is a side view showing a state in which the outboard motor main body **2** has been tilted up (a state in which the inclination angle is within a tilt range). The outboard motor main body **2** is turned around the tilt shaft **9** between a substantially vertical attitude and an attitude in which the outboard motor main body **2** is largely inclined while directing a front surface of the outboard motor main body **2** (front surface of the engine cover **19** and front surface of the casing **20**) toward the lower side as shown in FIG. 2. If the inclination angle of the outboard motor main body **2** when a lower end of the drive shaft **14** becomes closest to the hull **5a** is assumed as zero, the range within

which the inclination angle of the outboard motor main body **2** is small is a trim range, and the range within which the inclination angle of the outboard motor main body **2** is larger than an upper-limit boundary value of the trim range is a tilt range. In FIG. **2**, a state in which the inclination angle of the outboard motor main body **2** is a lower-limit boundary value of the trim range is shown by an alternate long and short dashed line, whereas a state in which the inclination angle of the outboard motor main body **2** is the upper-limit boundary value of the trim range is shown by an alternate long and two short dashed line. Additionally, in FIG. **2**, a state in which the inclination angle of the outboard motor main body **2** is an upper limit value of the tilt range (full-tilt-up) is shown by a solid line. The upper limit value of the tilt range is, for example, a maximum value of the inclination angle of the outboard motor main body **2**. The outboard motor main body **2** is able to be held at an arbitrary position of the trim range and of the tilt range.

(27) “Turning the outboard motor main body **2** upwardly within a trim range” is referred to as “trim up,” and “turning the outboard motor main body **2** downwardly within a trim range” is referred to as “trim down.” As a more functional definition, “turning the outboard motor main body **2** upwardly for a trim adjustment of the vessel” is referred to as “trim up,” and “turning the outboard motor main body **2** downwardly for a trim adjustment of the vessel” is referred to as “trim down.” On the other hand, “turning the outboard motor main body **2** upwardly with the aim of raising the propeller **16** over the surface of water” is referred to as “tilt up,” and “turning the outboard motor main body **2** downwardly with the aim of lowering the propeller **16** under the surface of water” is referred to as “tilt down.” Therefore, there is a case in which “tilt up” and “tilt down” are used for the up-and-down movement of the outboard motor main body **2** in both the trim range and the tilt range.

(28) FIG. **3** is a schematic view of the mount assembly **3**, and shows a state of a portion of the mount assembly **3** seen from the rear. The clamp bracket **6** includes a pair of clamp brackets **6** as shown in FIG. **3**. The pair of clamp brackets **6** are located at a distance from each other in the left-right direction. A portion of the swivel bracket **7** and the PTT mechanism **4** are located between the pair of clamp brackets **6**.

(29) The PTT mechanism **4** includes, for example, two trim cylinders **30** and a single tilt cylinder **40**. Each of the trim cylinders **30** and the tilt cylinder **40** are located between the two clamp brackets **6**. The two trim cylinders **30** are located so as to coincide with each other when seen from the left-right direction of the hull **5a**. The two trim cylinders **30** are located on both the right/left sides of the tilt cylinder **40**. Each of the trim cylinders **30** is obliquely located along the front-rear direction of the vessel so that an upper end of the trim cylinder **30** is placed at a more rearward position than a lower end of the trim cylinder **30**. Likewise, the tilt cylinder **40** is obliquely located along the front-rear direction of the vessel so that an upper end of the tilt cylinder **40** is placed at a more rearward position than a lower end of the tilt cylinder **40**. Each of the trim cylinders **30** and the tilt cylinder **40** are each, for example, a hydraulic cylinder. A tank **T1** in which hydraulic oil is stored and an electric motor **M1** that drives a hydraulic pump that supplies hydraulic oil are located between the two clamp brackets **6** as shown in FIG. **3**. The outboard motor main body **2** and the swivel bracket **7** are turned around the tilt shaft **9** by each of the trim cylinders **30** and the tilt cylinder **40**. The electric motor **M1** and the hydraulic pump are one example of an electrically-operated actuator in a preferred embodiment of the present invention, and supply a driving force to the PTT mechanism **4**.

(30) The cylinder main body **31** of each of the trim cylinders **30** is coupled to a corresponding one of the clamp brackets **6**. The trim rod **32** obliquely upwardly protrudes from an upper end portion of the cylinder main body **31** toward the rear. The trim rod **32** is reciprocated in an axial direction of the trim rod **32** by hydraulic pressure in the cylinder main body **31**. An upper end portion of each of the trim rods **32** comes into contact with the swivel bracket **7** in a state in which the inclination angle of the outboard motor main body **2** is within the trim range as shown by the alternate long and two short dashed line in FIG. **2**. Therefore, in this state, the outboard motor main body **2** is

supported from the front side by the two trim rods **32** through the swivel bracket **7**. Additionally, when the inclination angle of the outboard motor main body **2** is large and reaches the tilt range, the upper end portion of each of the trim rods **32** is separated from the swivel bracket **7**. Therefore, the support of the outboard motor main body **2** by the two trim rods **32** is released.

(31) The inclination angle of the outboard motor main body **2** when the trim rods **32** are in contact with the swivel bracket **7** in a maximum expansion state is the upper-limit boundary value of the trim range. In other words, the upper-limit boundary value of the trim range is defined by the upper limit value of the inclination angle that is changeable by driving the trim cylinder **30**. On the other hand, a minimum value of the inclination angle that can be taken by the outboard motor main body **2** when the trim rod **32** is in a minimum expansion state, i.e., in a contracted state is the lower-limit boundary value of the trim range.

(32) A lower end portion of the cylinder main body **41** of the tilt cylinder **40** is coupled to the clamp bracket **6**. The tilt rod **42** obliquely upwardly protrudes from an upper end portion of the cylinder main body **41** toward the rear. An upper end portion of the tilt rod **42** is coupled to the swivel bracket **7**. The tilt rod **42** is reciprocated in an axial direction of the tilt rod **42** by hydraulic pressure in the cylinder main body **41**. The upper end portion of the tilt rod **42** is coupled to the swivel bracket **7** even in a state in which the inclination angle of the outboard motor main body **2** is within either of the trim range and the tilt range. Therefore, the outboard motor main body **2** is supported by the tilt cylinder **40** even in a state in which the inclination angle of the outboard motor main body **2** is within either of the trim range and the tilt range.

(33) In a state in which the inclination angle of the outboard motor main body **2** is within the trim range, the outboard motor main body **2** is supported by the two trim cylinders **30** and the single tilt cylinder **40**. Additionally, in this state, the outboard motor main body **2** is turned upwardly and downwardly around the tilt shaft **9** by the two trim cylinders **30** and the single tilt cylinder **40**. The inclination angle of the outboard motor main body **2** is larger in proportion to an increase in the amount of protrusion of each of the trim rods **32** and the tilt rod **42**. Additionally, when the inclination angle of the outboard motor main body **2** is large and reaches the tilt range, the support of the outboard motor main body **2** by the two trim cylinders **30** is released, and the outboard motor main body **2** is supported by the single tilt cylinder **40**. In this state, the outboard motor main body **2** is turned upwardly and downwardly around the tilt shaft **9** by the single tilt cylinder **40**. The inclination angle of the outboard motor main body **2** becomes larger in proportion to an increase in the amount of protrusion of the tilt rod **42**. The inclination angle of the outboard motor main body **2** is changeable within a range from the lower-limit boundary value of the trim range to the upper limit value of the tilt range by the extension and contraction of the tilt rod **42**.

(34) FIG. **4** is a cross-sectional view along line IV-IV of FIG. **1**. The upper anti-vibration mount **12** includes a shaft portion **51**, an elastic portion **52**, an outer cylinder portion **53**, and a fastener **54** as shown in FIG. **4**. The shaft portion **51** is made of metal, such as aluminum, and is integral with the upper bracket **10**. A screw hole **51a** extending along a central axis of the shaft portion **51** is provided in the shaft portion **51**. The elastic portion **52** is provided in a circular cylindrical shape with an elastic material, such as rubber or sponge. The elastic portion **52** is coaxially fitted and attached to the shaft portion **51** in a state of surrounding the shaft portion **51**.

(35) The outer cylinder portion **53** has a circular cylindrical shape and is made of a metal material, such as aluminum. The outer cylinder portion **53** is coaxially fitted and attached to the elastic portion **52** in a state of surrounding the elastic portion **52**. The outer cylinder portion **53** is not in contact with the shaft portion **51**. The elastic portion **52** may be always compressed between the shaft portion **51** and the outer cylinder portion **53**. A portion of the outer cylinder portion **53** is housed in a concave portion **20a** that is hollowed from a surface of the casing **20**. The outer cylinder portion **53** is fixed to the casing **20** of the outboard motor main body **2** by a fixing member **24**. The fixing member **24** is fastened to the casing **20** by a fastening member **25**, such as a bolt. The outer cylinder portion **53** may be regarded as an element of the casing **20**, and not an element

of the anti-vibration mount **12**, because the outer cylinder portion **53** is fixed to the casing **20**. The fastening member **54** is, for example, a bolt screwed into a screw hole **51a** of the shaft portion **51**. Although not shown, the elastic portion **52** is clamped in the axial direction between a head portion of the bolt and a predetermined portion of the upper bracket **10**.

(36) The lower anti-vibration mount **12** has the same configuration as the upper anti-vibration mount **12** except that the shaft portion **51** of the lower anti-vibration mount **12** is integral with the lower bracket **11**. In each of the anti-vibration mounts **12**, the elastic portion **52** is interposed between the shaft portion **51** fixed to the swivel bracket **7** through the steering shaft **8** and the outer cylinder portion **53** fixed to the casing **20** of the outboard motor main body **2**, and is elastically deformable. Therefore, the outboard motor main body **2** is elastically supported by the anti-vibration mount **12**. Vibrations of the outboard motor main body **2** are attenuated by the elastic deformation of the elastic portion **52**, and thus are prevented from being transmitted to the hull **5a**.

(37) FIG. 5A is a schematic view of a main portion of the PTT mechanism **4**, and shows an example of the anti-vibration structure BS of the outboard motor. The tilt cylinder **40** defining the hydraulic cylinder includes the cylinder main body **41**, the tilt rod **42**, a piston **43**, a free piston **44**, a pair of oil chambers **45** and **46**, and a gas chamber **47** as shown in FIG. 5A.

(38) The cylinder main body **41** includes a circular cylindrical tube having a bottom. The cylinder main body **41** includes a pair of end portions **41a** and **41b**, an outer peripheral surface **41c**, an inner peripheral surface **41d**, a contact portion **41e**, a first port P1, and a second port P2. A portion of the tilt rod **42** is inserted in the cylinder main body **41**. The piston **43** is housed in the cylinder main body **41**, and defines the pair of oil chambers **45** and **46** in the cylinder main body **41**. The piston **43** slides in the axial direction of the tilt rod **42** along the inner peripheral surface **41d** of the cylinder main body **41** in the cylinder main body **41**.

(39) The tilt rod **42** extends through, for example, the upper oil chamber **46**, which is one of the pair of oil chambers **45** and **46**, and extends outwardly from the cylinder main body **41**. The tilt cylinder **40** is extendable and contractible in the expansion/contraction direction X corresponding to the axial direction of the tilt rod **42** in response to the movement in the axial direction of the tilt rod **42**. The first port P1 includes an opening that extends through the outer peripheral surface **41c** and the inner peripheral surface **41d** of the cylinder main body **41** and that communicates with the lower oil chamber **45**. The second port P2 includes an opening that extends through the outer peripheral surface **41c** and the inner peripheral surface **41d** of the cylinder main body **41** and that communicates with the upper oil chamber **46**.

(40) The free piston **44** is housed in the cylinder main body **41**. The free piston **44** is located on the side opposite to the tilt rod **42** with respect to the piston **43**, and defines the gas chamber **47** in the cylinder main body **41**. The free piston **44** is an example of a movable partition KS by which the lower oil chamber **45** and the gas chamber **47** are divided from each other in the cylinder main body **41**. The free piston **44** slides in the axial direction of the tilt rod **42** along the inner peripheral surface **41d** of the cylinder main body **41** in the cylinder main body **41** on the side opposite to the tilt rod **42**.

(41) The free piston **44** has a disk shape, and includes two end surfaces **44a**, **44b** and an outer peripheral surface **44c**. The upper end surface **44a** faces the lower oil chamber **45**. The lower end surface **44b** faces the gas chamber **47**. A seal **44e**, such as an O-ring, is held in a circumferential groove **44d** provided at the outer peripheral surface **44c**. The seal **44e** seals a portion between the outer peripheral surface **44c** of the free piston **44** and the inner peripheral surface **41d** of the cylinder main body **41**. A gas-filled structure GSS that defines the gas chamber **47** that is filled with gas is provided in the cylinder main body **41** by the free piston **44**. The tilt cylinder **40** is able to fulfill a function of a gas damper GD by the gas-filled structure GSS provided in the cylinder main body **41**. The gas with which the gas chamber **47** is filled may be air, or may be an inert gas, such as a nitrogen gas. An example of the anti-vibration structure BS of the outboard motor includes the tilt cylinder **40** and the gas damper GD.

(42) In response to the up-and-down movement of the free piston **44**, the volume of the gas chamber **47** changes and the inner pressure of the gas chamber **47** changes. The inner pressure of the gas chamber **47** generates a force that extends the tilt rod **42** through the free piston **44** and the piston **43**. In other words, the gas damper GD functions as a gas spring in a direction in which the tilt cylinder **40** is extended.

(43) The contact portion **41e** of the cylinder main body **41** is located in the gas chamber **47**. The contact portion **41e** includes a step that protrudes inwardly from the inner peripheral surface **41d** of the cylinder main body **41** and that faces the free piston **44**. The contact portion **41e** is in contact with the lower end surface **44b** of the free piston **44** when the free piston **44** moves to a lower stroke end as shown in FIG. 5B. The gas chamber **47** includes a first portion **47a** that is located at a higher position than the contact portion **41e** and a second portion **47b** that is located at a lower position than the contact portion **41e**. The position of the stroke end of the free piston **44** is regulated in a state in which the contact portion **41e** is in contact with the free piston **44**. Additionally, the gas chamber **47** is provided only with the second portion **47b** in a state in which the contact portion **41e** is in contact with the free piston **44**. In other words, the contact portion **41e** fulfills a function to regulate the minimum volume of the gas chamber **47**, and the minimum volume of the gas chamber **47** corresponds to the volume of the second portion **47b**.

(44) The PTT mechanism **4** includes a hydraulic circuit **60** that supplies/discharges hydraulic oil to the pair of oil chambers **45** and **46** of the tilt cylinder **40**. It is configured so that the tilt rod **42** is extended and contracted in response to the supply/discharge of hydraulic oil from the hydraulic circuit **60** to the pair of oil chambers **45** and **46**. The hydraulic circuit **60** includes a hydraulic pump **61**, a reservoir tank **62**, and a main valve assembly **63**. The hydraulic pump **61** is driven by the electric motor M1. The electric motor M1 is able to make normal rotation and reverse rotation, and the hydraulic pump **61** is normally rotationally or reversely rotationally driven by the electric motor M1. The main valve assembly **63** is connected to two ports **61a** and **61b** of the hydraulic pump **61**.

(45) The main valve assembly **63** includes a cylinder **64**, a shuttle **65** that slides in the cylinder **64**, and two on-off valves **66** and **67** located at both sides of the cylinder **64**, respectively. Two oil chambers **64a** and **64b** are defined at both sides of the shuttle **65** in the cylinder **64**. The two ports **61a** and **61b** of the hydraulic pump **61** are joined to the two oil chambers **64a** and **64b** through oil passages L11 and L12, respectively. The on-off valves **66** and **67** are check valves, and are opened by an increase in hydraulic pressure in the oil chambers **64a** and **64b** corresponding to the on-off valves **66** and **67**. Additionally, the on-off valves **66** and **67** are opened by being pressed by corresponding needles **65a** and **65b** joined to the shuttle **65**. In other words, a shuttle valve includes the shuttle **65** and the needles **65a** and **65b**.

(46) The lower oil chamber **45** of the tilt cylinder **40** is joined to the on-off valve **66** of the main valve assembly **63** through the first port P1 and an oil passage L1. The upper oil chamber **46** of the tilt cylinder **40** is joined to the on-off valve **67** of the main valve assembly **63** through the second port P2 and an oil passage L2. The two on-off valves **66** and **67** of the main valve assembly **63** are closed in a state in which the electric motor M1 has stopped running. Therefore, hydraulic oil is not supplied/discharged from the hydraulic circuit **60** to the oil chambers **45** and **46** of the tilt cylinder **40**. The gas-filled structure GSS provided in the tilt cylinder **40** functions as a gas damper GD in a state in which hydraulic oil is not supplied/discharged from the hydraulic circuit **60** to the oil chambers **45** and **46** of the tilt cylinder **40**.

(47) When the hydraulic pump **61** is normally rotationally driven by the electric motor M1, the hydraulic pump **61** receives hydraulic oil from the port **61b**, and discharges hydraulic oil from the port **61a**. The discharged hydraulic oil is supplied from the main valve assembly **63** to the oil passage L1. As a result, the hydraulic oil is supplied to the lower oil chamber **45** of the tilt cylinder **40** through the first port P1. Thus, the tilt rod **42** is extended, and turns the swivel bracket **7** upwardly. Hydraulic oil is replenished from the reservoir tank **62** through a one-way valve **68** when hydraulic oil is insufficient. On the other hand, hydraulic oil in the upper oil chamber **46** of the tilt

cylinder **40** is drawn into the hydraulic pump **61** through the second port **P2**, the oil passage **L2**, and the main valve assembly **63**. At this time, the on-off valve **66** of the main valve assembly **63** is pressed and opened by the needle **65b** of the shuttle **65**.

(48) When the hydraulic pump **61** is reversely rotationally driven by the electric motor **M1**, the hydraulic pump **61** receives hydraulic oil from the port **61a**, and discharges hydraulic oil from the port **61b**. The discharged hydraulic oil is supplied from the main valve assembly **63** to the oil passage **L2**. As a result, hydraulic oil is supplied to the upper oil chamber **46** of the tilt cylinder **40** through the second port **P2**. As a result, the tilt rod **42** is contracted, and turns the swivel bracket **7** downwardly. Hydraulic oil is replenished from the reservoir tank **62** through a one-way valve **69** when hydraulic oil is insufficient. On the other hand, hydraulic oil in the lower oil chamber **45** of the tilt cylinder **40** is drawn into the hydraulic pump **61** through the first port **P1**, the oil passage **L1**, and the main valve assembly **63**. The tilt rod **42** is contracted by being pushed by the swivel bracket **7**.

(49) According to the present preferred embodiment, the outboard motor **1** includes the tilt cylinder **40** as an example of a hydraulic cylinder that turns the swivel bracket **7** around the tilt shaft **9** with respect to the clamp bracket **6** as shown in FIG. **1**. Additionally, the outboard motor **1** includes the gas damper **GD** that attenuates an external force that acts in the extension/contraction direction **X** of the tilt cylinder **40** as shown in FIG. **5A**. Therefore, an external force acting in the extension/contraction direction **X** of the tilt cylinder **40** is attenuated by the gas damper **GD**. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull **5a**, and makes it possible to improve the riding comfort of the vessel.

(50) Additionally, the gas damper **GD** includes the gas-filled structure **GSS** in the cylinder main body **41** of the tilt cylinder **40** and that is filled with gas. The gas-filled structure **GSS** in the cylinder main body **41** functions as a damper, and, as a result, an external force that acts in the extension/contraction direction **X** of the tilt cylinder **40** is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull **5a**.

(51) Additionally, the gas-filled structure **GSS** includes the free piston **44** that slides in the cylinder main body **41** and that defines the gas chamber **47** in the cylinder main body **41**. Gas with which the gas chamber **47** defined by the free piston **44** in the cylinder main body **41** is filled functions as a damper, and, as a result, an external force that acts in the extension/contraction direction **X** of the tilt cylinder **40** is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull **5a**.

(52) Additionally, the cylinder main body **41** includes the contact portion **41e** contactable with the free piston **44** so as to regulate the position of a stroke end of the free piston **44** as shown in FIG. **5B**. The minimum volume of the gas chamber **47** is regulated in a state in which the contact portion **41e** is in contact with the free piston **44**. Therefore, an increase of the pressure in the gas chamber **47** is regulated.

(53) Additionally, the mount assembly **3** includes the anti-vibration mount **12** that elastically supports the outboard motor main body **2** with respect to the swivel bracket **7** as shown in FIG. **1**. The gas damper **GD** functions as a gas spring having a spring constant lower than that of the anti-vibration mount **12**. Therefore, it becomes possible to extend/contract the gas damper **GD** even when an external force that acts in the extension/contraction direction **X** of a hydraulic cylinder (for example, the tilt cylinder **40**) is small. Therefore, it is possible to improve an anti-vibration effect, and it is possible to further significantly reduce or prevent vibrations transmitted to the hull **5a**. Particularly, the gas damper **GD** that functions as a gas spring having a spring constant lower than that of the anti-vibration mount **12** makes it possible to improve an anti-vibration effect when the engine rotates at a low speed during low-speed navigation, such as trolling.

(54) FIG. **6A** is a schematic view shown to describe another example of the anti-vibration structure **BS** in the outboard motor **1**. The example of FIG. **6A** mainly differs from the example of FIG. **5A** as follows. Specifically, the gas damper **GD** includes a sub-cylinder **70** hydraulically connected to,

for example, an oil chamber 45 that is one of the oil chambers 45 of the tilt cylinder 40 and that is an example of a hydraulic cylinder. The sub-cylinder 70 includes a sub-cylinder main body 71, a sub-oil chamber 72, a gas chamber 73, and a free piston 74. The sub-cylinder main body 71 includes a circular cylindrical tube. The cylinder main body 41 includes a pair of end portions 71a, 71b, an outer peripheral surface 71c, an inner peripheral surface 71d, and a contact portion 71e. (55) The sub-oil chamber 72 and the gas chamber 73 are located in the sub-cylinder main body 71. The free piston 74 is an example of a movable partition KS that is housed in the sub-cylinder main body 71 and by which the sub-oil chamber 72 and the gas chamber 73 are divided from each other. The free piston 74 slides in the sub-cylinder main body 71. The sub-oil chamber 72 is located at a side of one end portion 71a with respect to the free piston 74. The gas chamber 73 is located at the other side of an end portion 71b with respect to the free piston 74. An oil passage L3 is provided between the tilt cylinder 40 and the sub-cylinder 70. The oil chamber 45 of the tilt cylinder 40 and the sub-oil chamber 72 of the sub-cylinder 70 are able to communicate with each other through the oil passage L3.

(56) Two ports P3 and P4 are provided at two end portions of the oil passage L3, respectively. The port P3 is located at, for example, the end portion 41b of the cylinder main body 41 of the tilt cylinder 40. The port P4 is located at the end portion 71a of the sub-cylinder main body 71. For example, an on-off valve V1 that is manually operable is interposed in the oil passage L3. The on-off valve V1 is able to interrupt the flow of hydraulic oil between the oil chamber 45 of the tilt cylinder 40 and the sub-oil chamber 72 of the sub-cylinder 70 in the closed state. The on-off valve V1 allows the flow of hydraulic oil between the oil chamber 45 and the sub-oil chamber 72 in an opened state.

(57) The free piston 74 is the same in configuration as the free piston 44 of FIG. 5A, and includes two end surfaces 74a, 74b, an outer peripheral surface 74c, and a circumferential groove 74d as shown in FIG. 6B. The end surface 74a, which is one of the two end surfaces, faces the sub-oil chamber 72, and the other end surface 74b faces the gas chamber 73. A seal 74e, such as an O-ring, is held in the circumferential groove 74d. The seal 74e seals a portion between the outer peripheral surface 74c of the free piston 74 and the inner peripheral surface 71d of the sub-cylinder main body 71. A gas-filled structure GSS that defines the gas chamber 73 filled with gas is provided in the sub-cylinder main body 71 by the free piston 74. The sub-cylinder 70 functions as a gas damper GD.

(58) The free piston 74 rises and falls when the tilt cylinder 40 extends and contracts. In response to the up-and-down movement of the free piston 74, the volume of the gas chamber 73 changes and the inner pressure of the gas chamber 73 changes. The inner pressure of the gas chamber 73 generates a force that extends the tilt rod 42 through the free piston 74 and the piston 43. A force that extends the tilt rod 42 increases when the inner pressure of the gas chamber 73 increases in response to the contraction of the tilt cylinder 40. In other words, the gas damper GD functions as a gas spring that resists the contraction of the tilt cylinder 40. Preferably, the gas damper GD functions as a gas spring having a spring constant lower than that of the anti-vibration mount 12.

(59) The contact portion 71e of the sub-cylinder main body 71 is located in the gas chamber 73. The contact portion 71e includes a step that protrudes inwardly from the inner peripheral surface 71d of the sub-cylinder main body 71 and that faces a side of the free piston 74. When the free piston 74 moves to the lower stroke end, the contact portion 71e is in contact with the other end surface 74b of the free piston 74 as shown in FIG. 6B. The gas chamber 73 includes a first portion 73a located higher than the contact portion 71e and a second portion 73b located lower than the contact portion 71e. The position of the stroke end of the free piston 74 is regulated in a state in which the contact portion 71e is in contact with the free piston 74. Additionally, the gas chamber 73 is provided only with the second portion 73b in a state in which the contact portion 71e is in contact with the free piston 74. In other words, the contact portion 71e fulfills a function to regulate the minimum volume of the gas chamber 73, and this minimum volume of the gas chamber 73

corresponds to the volume of the second portion **73b**.

(60) According to the present preferred embodiment, the sub-cylinder **70** hydraulically connected to the oil chamber **45** of the tilt cylinder **40** functions as a gas damper GD, and, as a result, an external force that acts in the extension/contraction direction X of the tilt cylinder **40** is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(61) Additionally, in the sub-cylinder main body **71**, the sub-oil chamber **72** and the gas chamber **73** are partitioned by the free piston **74**. The sub-oil chamber **72** is in communication with the oil chamber **45** of the tilt cylinder **40**. Gas with which the gas chamber **73** is filled functions as a damper, and, as a result, an external force that acts in the extension/contraction direction X of the tilt cylinder **40** is attenuated. Therefore, it is possible to significantly reduce or prevent vibrations transmitted to the hull.

(62) Additionally, the sub-cylinder main body **71** includes the contact portion **71e** that is contactable with the free piston **74** so as to regulate the position of the stroke end of the free piston **74**. The contact portion **71e** regulates the minimum volume of the gas chamber **73** in a state of being in contact with the free piston **74**. Therefore, an increase of the pressure in the gas chamber **73** is regulated.

(63) Additionally, the on-off valve **V1** that interrupts the flow of hydraulic oil between the oil chamber **45** of the tilt cylinder **40** and the sub-oil chamber **72** of the sub-cylinder **70** is able to be manually operated. The sub-cylinder **70** functions as a gas damper GD in a state in which the on-off valve **V1** has been opened. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull. For example, during low-speed navigation, such as trolling, it is possible to improve an anti-vibration effect when the engine rotates at a low speed in a state in which the on-off valve **V1** has been opened. It should be noted that the on-off valve **V1** is manually closed when the navigation speed of the vessel increases.

(64) Additionally, the gas damper GD functions as a gas spring having a spring constant lower than that of the anti-vibration mount **12**, thus making it possible to improve an anti-vibration effect when the engine rotates at a low speed during low-speed navigation, such as trolling.

(65) FIG. **7** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The example of FIG. **7** mainly differs from the example of FIG. **6A** as follows. Specifically, an electromagnetic control valve **V2** is provided as an on-off valve interposed in the oil passage **L3** by which the oil chamber **45** of the tilt cylinder **40** and the sub-oil chamber **72** of the sub-cylinder **70** are in communication with each other. The ECU **18**, which is an example of a controller, calculates an engine rotation speed of the engine **13**, which is an example of a prime mover, based on a detection signal of the rotation speed detector **23**. The ECU **18** is configured or programmed to shut off the control valve **V2**, which is an on-off valve, when the detected rotation speed exceeds a predetermined threshold value. In other words, when the detected engine rotation speed is equal to or below the predetermined threshold value, the control valve **V2** is opened.

(66) According to the present preferred embodiment, the same effect as the example of FIG. **6A** is achieved. Additionally, when the detection value of the engine rotation speed is lower than the threshold value, the sub-cylinder **70** is able to automatically function as a gas damper GD. This makes it possible to automatically improve an anti-vibration effect when the engine rotates at a low speed during low-speed navigation, such as trolling. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull.

(67) FIG. **8** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The example of FIG. **8** mainly differs from the example of FIG. **7** as follows. Specifically, a sub-cylinder **70P** hydraulically connected to the upper oil chamber **46** of the tilt cylinder **40** is provided. The sub-cylinder **70P** has the same in configuration as the sub-cylinder **70** hydraulically connected to the lower oil chamber **45** of the tilt cylinder **40**. The elements of the sub-cylinder **70P** are provided with the same reference characters as the

equivalent elements of the sub-cylinder **70**. The upper oil chamber **46** of the tilt cylinder **40** and the sub-oil chamber **72** of the sub-cylinder **70P** are in communication with each other through an oil passage **L4**. Two ports **P5** and **P6** are respectively located at two end portions of the oil passage **L4**. The port **P5** is located at, for example, the end portion **41a** of the cylinder main body **41** of the tilt cylinder **40**. The port **P6** is located at the end portion **71a** of the sub-cylinder main body **71** of the sub-cylinder **70P**.

(68) An electromagnetic control valve **V3**, which is an on-off valve, interrupts the flow of hydraulic oil between the upper oil chamber **46** of the tilt cylinder **40** and the sub-oil chamber **72** of the sub-cylinder **70P**. The control valve **V3** is interposed in the oil passage **L4**. The ECU **18** is configured or programmed to shut off the control valve **V2** and the control valve **V3** when the engine rotation speed detected by the rotation speed detector **23** exceeds a predetermined threshold value. When the detected engine rotation speed is equal to or below the predetermined threshold value, the control valve **V2** and the control valve **V3** are opened.

(69) When the tilt cylinder **40** that has received an external force in the extension/contraction direction **X** is extended in a state in which the control valve **V2** and the control valve **V3** have been opened, hydraulic oil flows from the upper oil chamber **46** to the sub-oil chamber **72** of the sub-cylinder **70P**, and gas in the gas chamber **73** of the sub-cylinder **70P** is compressed. Additionally, when the tilt cylinder **40** that has received an external force in the extension/contraction direction **X** is contracted, hydraulic oil flows from the lower oil chamber **45** to the sub-oil chamber **72** of the sub-cylinder **70**, and gas in the gas chamber **73** of the sub-cylinder **70** is compressed.

(70) According to this structural configuration, the same effect as the example of FIG. 7 is achieved. Additionally, each of the sub-cylinders **70P** and **70** that respectively correspond to the extension and the contraction of the tilt cylinder **40** functions as a gas damper **GD**. This makes it possible to attenuate an extension/contraction bidirectional external force that acts on the tilt cylinder **40**. Therefore, the attenuation effect is high. This makes it possible to further significantly reduce or prevent vibrations transmitted to the hull.

(71) FIG. 9 is a schematic view shown to describe still another example of the anti-vibration structure **BS** in the outboard motor **1**. The example of FIG. 9 mainly differs from the example of FIG. 8 as follows. Specifically, a single (only one) sub-cylinder **80** hydraulically connected to the two oil chambers **45** and **46** of the tilt cylinder **40** is provided. The sub-cylinder **80** includes two sub-oil chambers **81** and **82** and two gas chambers **83** and **84**. The lower oil chamber **45** of the tilt cylinder **40** and the lower sub-oil chamber **81** of the sub-cylinder **80** are in communication with each other through the oil passage **L3**. The electromagnetic control valve **V2**, which is an on-off valve, is located in the oil passage **L3**. The upper oil chamber **46** of the tilt cylinder **40** and the upper sub-oil chamber **82** of the sub-cylinder **80** are in communication with each other through the oil passage **L4**.

(72) The sub-cylinder **80** includes a sub-cylinder main body **85** and a rod piston **86** that functions as a movable partition **KS**. The rod piston **86** includes a gas piston **89** and two rods **87** and **88** that extend from a central portion **89H** of the gas piston **89**. The rod piston **86** provides an all-in-one piston **HGP** that functions as a hydraulic piston and as a gas piston. The gas piston **89** divides the two gas chambers **83** and **84** from each other. The rod piston **86** divides the two sub-oil chambers **81** and **82** from each other. The lower rod **87** defines a portion of the lower sub-oil chamber **81**, and functions as a hydraulic piston. The upper rod **88** defines a portion of the upper sub-oil chamber **82**, and functions as a hydraulic piston.

(73) The sub-cylinder main body **85** includes a circular cylindrical center tube **90**, two circular cylindrical end tubes **91** and **92**, and two end covers **93** and **94**. The two end tubes **91** and **92** are fitted and fixed to two end portions **90a** and **90b** of the center tube **90**. The inner diameter of each of the two end tubes **91** and **92** is smaller than the inner diameter of the center tube **90**. The two end covers **93** and **94** cover end portions of the two end tubes **91** and **92**.

(74) The gas piston **89** has a disk shape, and is housed in the center tube **90**. A gas cylinder is

provided with the center tube **90** and the gas piston **89**. The gas piston **89** includes two end surfaces **89a**, **89b** and an outer peripheral surface **89c**. The gas piston **89** slides on an inner peripheral surface **90c** of the center tube **90**. A seal **89e**, such as an O-ring, is housed in a circumferential groove **89d** provided at the outer peripheral surface **89c** of the gas piston **89**. A portion between the outer peripheral surface **89c** of the gas piston **89** and the inner peripheral surface **90c** of the center tube **90** is sealed by the seal **89e**.

(75) The two rods **87** and **88** of the rod piston **86** functioning as a hydraulic piston extend from a central portion **89H** of the gas piston **89** toward mutually opposite sides along a central axis of the center tube **90**. The two rods **87** and **88** are inserted in the corresponding end tubes **91** and **92**. Each of the rods **87** and **88** is slidable along an inner peripheral surface of a corresponding one of the end tubes **91** and **92**. A hydraulic cylinder includes each of the rods **87** and **88** and a corresponding one of the end tubes **91** and **92**.

(76) The lower rod **87** includes an outer peripheral surface **87a**, a step **87b**, and an axial hole **87c**. The step **87b** is located at the outer peripheral surface **87a**. The axial hole **87c** has a bottom, and is in communication with the lower sub-oil chamber **81**. In some cases, the axial hole **87c** is not provided. The upper rod **88** includes an outer peripheral surface **88a**, a step **88b**, and an axial hole **88c**. The step **88b** is provided at the outer peripheral surface **88a**. The axial hole **88c** has a bottom, and is in communication with the upper sub-oil chamber **82**. In some cases, the axial hole **88c** is not provided.

(77) The lower sub-oil chamber **81** is defined by the lower rod **87** and by the lower end cover **93** in the lower end tube **91**. The lower end cover **93** includes a communication hole **93a** through which the lower sub-oil chamber **81** and the oil passage **L3** are in communication with each other. The port **P4** of the end portion of the oil passage **L3** is located at an end portion of the communication hole **93a**. The upper sub-oil chamber **82** is defined by the upper rod **88** and by the upper end cover **94** in the upper end tube **92**. The upper end cover **94** includes a communication hole **94a** through which the upper sub-oil chamber **82** and the oil passage **L4** are in communication with each other. The port **P6** of the end portion of the oil passage **L4** is located at an end portion of the communication hole **94a**.

(78) The lower gas chamber **83** is surrounded by the outer peripheral surface **87a** of the lower rod **87**, the inner peripheral surface **90c** of the center tube **90**, the lower end surface **89a** of the gas piston **89**, and an end surface **91a** of the lower end tube **91**. Thus, a gas-filled structure GSS is provided. The upper gas chamber **84** is surrounded by the outer peripheral surface **88a** of the upper rod **88**, the inner peripheral surface **90c** of the center tube **90**, the upper end surface **89b** of the gas piston **89**, and an end surface **92a** of the upper end tube **92**. Thus, a gas-filled structure GSS is provided.

(79) When the tilt cylinder **40** that has received an external force in the extension/contraction direction **X** is contracted in a state in which the control valve **V2** has been opened, hydraulic oil flows from the lower oil chamber **45** of the tilt cylinder **40** into the lower sub-oil chamber **81** of the sub-cylinder **80** through the oil passage **L3**. Therefore, the rod piston **86** and the gas piston **89** rise, and gas in the upper gas chamber **84** is compressed. Additionally, when the tilt cylinder **40** that has received an external force in the extension/contraction direction **X** is extended, hydraulic oil flows from the upper oil chamber **46** of the tilt cylinder **40** into the upper sub-oil chamber **82** of the sub-cylinder **80**. Therefore, the rod piston **86** and the gas piston **89** fall, and gas in the lower gas chamber **83** is compressed.

(80) According to this structural configuration, the single sub-cylinder **80** having gas-filled structures GSS that respectively correspond to the extension and the contraction of the tilt cylinder **40** functions as an extension/contraction bidirectional gas damper **GD**. Therefore, the use of the single sub-cylinder **80** makes it possible to attenuate an extension/contraction bidirectional external force that acts on the tilt cylinder **40**. Therefore, the attenuation effect is high. This makes it possible to further significantly reduce or prevent vibrations transmitted to the hull.

(81) Additionally, the step **87b** of the lower rod **87** is in contact with the end surface **91a** of the lower end tube **91** that is an example of the contact portion, and, as a result, the minimum volume of the lower gas chamber **83** is regulated. Additionally, the step **88b** of the upper rod **88** is in contact with the end surface **92a** of the upper end tube **92** that is an example of the contact portion, and, as a result, the minimum volume of the upper gas chamber **84** is regulated. This makes it possible to control an increase of the inner pressure in each of the gas chambers **83** and **84**.

(82) FIG. **10** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The example of FIG. **10** mainly differs from the example of FIG. **7** as follows. Specifically, the upper oil chamber **46** of the tilt cylinder **40** is in communication with the sub-oil chamber **72** of the sub-cylinder **70** through an oil passage L5. The two oil chambers **45** and **46** of the tilt cylinder **40** are both in communication with the sub-oil chamber **72** of the sub-cylinder **70**. A port P7, which is one of two ports P7, P8 of end portions of the oil passage L5, is located at the end portion **41a** of the cylinder main body **41** of the tilt cylinder **40**. The other port P8 is located at the end portion **71a** of the sub-cylinder main body **71**.

(83) When the tilt cylinder **40** that has received an external force in the extension/contraction direction X is contracted in a state in which the control valve V2 has been opened, hydraulic oil flows from the lower oil chamber **45** of the tilt cylinder **40** into the sub-oil chamber **72** through the oil passage L3. When the tilt cylinder **40** is extended, hydraulic oil flows from the upper oil chamber **46** of the tilt cylinder **40** into the sub-oil chamber **72** through the oil passage L5. Either when the tilt cylinder **40** is extended or when the tilt cylinder **40** is contracted, the free piston **74** falls, and gas in the gas chamber **73** is compressed.

(84) According to this structural configuration, the single sub-cylinder **70** functions as an extension/contraction bidirectional gas damper GD. Therefore, the use of the single sub-cylinder **70** makes it possible to attenuate an extension/contraction bidirectional external force that acts on the tilt cylinder **40**. Therefore, the attenuation effect is high. This makes it possible to further significantly reduce or prevent vibrations transmitted to the hull.

(85) FIG. **11** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The example of FIG. **11** mainly differs from the example of FIG. **6A** as follows. Specifically, the gas damper GD includes a gas cylinder **100**. The gas cylinder **100** slidably houses the end portion **41b** of the cylinder main body **41** of the tilt cylinder **40** on the side opposite to the tilt rod **42**. The gas cylinder **100** defines a gas chamber **101**, filled with gas, with the end portion **41b** of the cylinder main body **41** of the tilt cylinder **40**. When the tilt cylinder **40** receives an external force in the extension/contraction direction X, the cylinder main body **41** of the tilt cylinder **40** slides in the gas cylinder **100**, and the inner pressure of gas in the gas chamber **101** changes. The gas cylinder **100** includes a circular cylindrical cylinder tube **102**, a bottom **103** that closes one end of the cylinder tube **102**, and a contact portion **104**. The bottom **103** of the gas cylinder **100** is coupled to a clamp bracket.

(86) The contact portion **104** is located in the gas chamber **101**. A seal **106**, such as an O-ring, is interposed between an inner peripheral surface **105** of the cylinder tube **102** and the outer peripheral surface **41c** of the cylinder main body **41**. The seal **106** is held in a circumferential groove provided at, for example, the inner peripheral surface of the cylinder tube **102**. The contact portion **104** may include a step that protrudes inwardly from the inner peripheral surface **105** of the cylinder tube **102** and that faces the end portion **41b** side of the cylinder main body **41** of the tilt cylinder **40**. The gas chamber **101** includes a first portion **101a** that is located at a higher position than the contact portion **104** and a second portion **101b** that is located at a lower position than the contact portion **104**. The gas chamber **101** is provided only with the second portion **101b** in a state in which the contact portion **104** is in contact with the end portion **41b** of the cylinder main body **41** of the tilt cylinder **40**. In other words, the contact portion **104** fulfills a function to regulate the minimum volume of the gas chamber **101**, and the minimum volume of the gas chamber **101** corresponds to the volume of the second portion **101b**.

(87) According to this structural configuration, the gas cylinder **100** that slidably houses the end portion **41b** of the cylinder main body **41** of the tilt cylinder **40** on the side opposite to the tilt rod **42** functions as a gas damper GD, and, as a result, an external force that acts in the extension/contraction direction X of the tilt cylinder **40** is attenuated. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull.

(88) Additionally, the contact portion **104** of the gas cylinder **100** is contactable with the end portion **41b** of the cylinder main body **41**, and, as a result, the position of the stroke end of the end portion **41b** of the cylinder main body **41** with respect to the gas cylinder **100** is regulated. Thus, the minimum volume of the gas chamber **101** is regulated, and an increase of the pressure in the gas chamber **101** is regulated.

(89) FIG. **12** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The example of FIG. **12** mainly differs from the example of FIG. **5A** as follows. Specifically, a hollow piston **110** is included as a piston of the tilt cylinder **40**. The hollow piston **110** includes a cylindrical inner peripheral surface **111** that defines a gas chamber **120** that is filled with gas. In other words, the hollow piston **110** includes a gas-filled structure GSS. The gas damper GD includes a double piston structure DPS including the hollow piston **110** and an internal piston **130**.

(90) The hollow piston **110** includes two end walls **112**, **113** and a peripheral sidewall **114**. A gas chamber **120** is defined in the hollow piston **110** by being surrounded by the two end walls **112**, **113** and the peripheral sidewall **114**. The inner peripheral surface **111** of the hollow piston **110** includes an inner peripheral surface of the peripheral sidewall **114**. The tilt rod **42** is inserted through the upper end wall **113**, and is coupled to the internal piston **130**. The hollow piston **110** includes a contact portion **115**.

(91) The internal piston **130** is housed in the hollow piston **110** so as to be slidable on the inner peripheral surface **111** of the hollow piston **110**. The internal piston **130** coupled to the tilt rod **42** partitions the gas chamber **120** into two gas chambers **121**, **122**. The internal piston **130** includes two end surfaces **131**, **132**, an outer peripheral surface **133**, and a seal **134** such as an O-ring. The lower gas chamber **121** is located between the lower end surface **131** of the internal piston **130** and the lower end wall **112** of the hollow piston **110**. The upper gas chamber **122** is located between the upper end surface **132** of the internal piston **130** and the upper end wall **113** of the hollow piston **110**. The seal **134** is held by a circumferential groove provided at the outer peripheral surface **133** of the internal piston **130**, and seals a portion between the outer peripheral surface **133** of the internal piston **130** and the inner peripheral surface **111** of the hollow piston **110**. The tilt cylinder **40** that has received an external force in the extension/contraction direction X is extended and contracted in a state in which the supply/discharge of hydraulic oil from the hydraulic circuit **60** to the oil chambers **45** and **46** of the tilt cylinder **40** have been stopped. In accordance with the extension and contraction, a corresponding one of the gas chambers **121**, **122** is compressed, and the function as a gas damper GD is achieved.

(92) The contact portion **115** includes a step that protrudes inwardly from the inner peripheral surface **111** of the hollow piston **110** in the lower gas chamber **121** and that faces the internal piston **130** side. When the internal piston **130** moves to the lower stroke end in the hollow piston **110**, the contact portion **115** is in contact with the internal piston **130**. The lower gas chamber **121** includes a first portion **121a** that is located at a higher position than the contact portion **115** and a second portion **121b** that is located at a lower position than the contact portion **115**. Although not shown, the position of the stroke end of the internal piston **130** is regulated in a state in which the contact portion **115** is in contact with the internal piston **130**. Additionally, the lower gas chamber **121** is provided only with the second portion **121b** in a state in which the contact portion **115** is in contact with the internal piston **130**. In other words, the contact portion **115** fulfills a function to regulate the minimum volume of the gas chamber **121**, and the minimum volume of the gas chamber **121** corresponds to the volume of the second portion **121b**.

(93) According to this structural configuration, the internal piston **130** performs a stroke in the hollow piston **110** in response to the extension/contraction of the tilt rod **42**, and, as a result, gas in a corresponding one of the gas chambers **121**, **122** in the hollow piston **110** is compressed. Thus, an external force that acts in the extension/contraction direction X of the tilt cylinder **40** is attenuated. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull.

(94) Additionally, the contact portion **115** of the hollow piston **110** is in contact with the internal piston **130**, and, as a result, the position of the stroke end of the internal piston **130** with respect to the hollow piston **110** is regulated. Thus, the minimum volume of the gas chamber **121** is regulated, and an increase of the pressure in the gas chamber **121** is regulated.

(95) Although not shown, a contact portion may be added in the upper gas chamber **122**. The contact portion is in contact with the internal piston **130**, and regulates the minimum volume of the upper gas chamber **122** when the internal piston **130** moves to the upper stroke end in the hollow piston **110**. Thus, an increase of the pressure in the gas chamber **122** is regulated.

(96) FIG. **13** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The gas damper GD includes two independent cellular foams **141**, **142** located in the two oil chambers **45** and **46**, respectively. The independent cellular foams **141**, **142** provide a gas-filled structure GSS in the cylinder main body **41** of the tilt cylinder **40**. The independent cellular foams **141**, **142** are made of resin foam or foamed rubber. According to this structural configuration, an external force that acts in the extension/contraction direction X of the tilt cylinder **40** is attenuated by the independent cellular foams **141**, **142** located in the two oil chambers **45** and **46**. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull.

(97) FIG. **14A** is a schematic view shown to describe still another example of the anti-vibration structure BS in the outboard motor **1**. The example of FIG. **14A** mainly differs from the example of FIG. **5A** as follows. Specifically, the gas damper GD includes a movable partition KS that defines the gas chamber **47** in the cylinder main body **41** of the tilt cylinder **40**. The movable partition KS includes, for example, a membrane **150**. The membrane **150** is made of, for example, rubber having elasticity or a metal sheet. The membrane **150** has, for example, a disk shape, and an outer peripheral edge **151** of the membrane **150** is fixed to a step **41f** of the inner peripheral surface **41d** of the cylinder main body **41** by an annular fixing member **160**. The gas chamber **47** is defined between the membrane **150** and the end portion **41b**, which corresponds to a bottom, of the cylinder main body **41**. The membrane **150** divides the oil chamber **45** and the gas chamber **47** from each other.

(98) A regulation portion R1 is provided that regulates a displacement amount when the membrane **150** is displaced so as to protrude toward the side of the gas chamber **47** by elastic deformation. The regulation portion R1 is provided with, for example, a convex portion **41g** that protrudes from the end portion **41b** of the cylinder main body **41** toward the central portion side of the membrane **150** in the gas chamber **47**. According to this structural configuration, gas in the gas chamber **47** defined by the movable partition KS in the tilt cylinder **40** functions as a damper, and, as a result, an external force that acts in the extension/contraction direction X of the tilt cylinder **40** is attenuated. This makes it possible to significantly reduce or prevent vibrations transmitted to the hull.

(99) Additionally, the displacement amount of the membrane **150** functioning as the movable partition KS is regulated by the convex portion **41g** functioning as the regulation portion R1. Thus, the minimum volume of the gas chamber **47** is regulated. Therefore, an increase of the pressure in the gas chamber **47** is regulated. The movable partition KS may be a diaphragm having elasticity.

(100) In each of the examples described above, as the hydraulic cylinder HC included in the anti-vibration structure BS, the trim cylinder **30** may be used instead of the tilt cylinder **40**, or both the tilt cylinder **40** and the trim cylinder **30** may be used. Additionally, in each of the examples of FIG. **8**, FIG. **9**, and FIG. **10**, the manually-operable on-off valve V1 may be used instead of the control

valves V2 and V3. Additionally, preferably, the spring constant when the gas damper GD functions as a gas spring is lower than the spring constant of the anti-vibration mount 12 in each of the examples of FIG. 7 to FIG. 14A.

(101) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An outboard motor comprising: an outboard motor main body; a mount assembly to mount the outboard motor main body to a hull, the mount assembly including a clamp bracket to be fixed to the hull and a swivel bracket that is coupled to the clamp bracket through a tilt shaft, turnable around the tilt shaft, and supports the outboard motor main body, the mount assembly further including an elastic mount to elastically support the outboard motor main body with respect to the swivel bracket; a hydraulic cylinder between the clamp bracket and the swivel bracket to turn the swivel bracket around the tilt shaft with respect to the clamp bracket; and a gas damper to damp an external force that acts in a telescopic direction of the hydraulic cylinder and to function as a gas spring having a spring constant lower than a spring constant of the elastic mount.
2. The outboard motor according to claim 1, wherein the hydraulic cylinder includes a cylinder main body, a piston slidable in the cylinder main body and defining a pair of oil chambers in the cylinder main body, and a rod extending outwardly from the cylinder main body through one of the pair of oil chambers, and is operable to extend and contract the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers; and the gas damper is inside the cylinder main body and filled with gas.
3. The outboard motor according to claim 2, wherein the gas damper includes a free piston slidable in the cylinder main body on a side opposite to the rod and defines a gas chamber filled with the gas in the cylinder main body.
4. The outboard motor according to claim 3, wherein the cylinder main body includes a contact portion contactable with the free piston to regulate a minimum volume of the gas chamber by regulating a position of a stroke end of the free piston.
5. An outboard motor comprising: an outboard motor main body; a mount assembly to mount the outboard motor main body to a hull, the mount assembly including a clamp bracket to be fixed to the hull and a swivel bracket that is coupled to the clamp bracket through a tilt shaft, turnable around the tilt shaft, and supports the outboard motor main body; a hydraulic cylinder between the clamp bracket and the swivel bracket to turn the swivel bracket around the tilt shaft with respect to the clamp bracket; and a gas damper to damp an external force that acts in a telescopic direction of the hydraulic cylinder; wherein the hydraulic cylinder includes a cylinder main body, a piston slidable in the cylinder main body and defining a pair of oil chambers in the cylinder main body, and a rod extending outwardly from the cylinder main body through one of the pair of oil chambers, and is operable to extend and contract the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers; and the gas damper includes a sub-cylinder hydraulically connected to at least one of the pair of oil chambers.
6. The outboard motor according to claim 5, wherein the sub-cylinder includes a sub-cylinder main body and a free piston slidable in the sub-cylinder main body that divides a sub-oil chamber in communication with the at least one of the pair of oil chambers of the hydraulic cylinder and a gas chamber filled with the gas from each other.
7. The outboard motor according to claim 6, wherein the sub-cylinder main body includes a contact portion contactable with the free piston to regulate a minimum volume of the gas chamber by regulating a position of a stroke end of the free piston.

8. The outboard motor according to claim 6, wherein the sub-cylinder includes a pair of sub-cylinders, and sub-oil chambers of the pair of sub-cylinders are connected to the pair of oil chambers of the hydraulic cylinder, respectively.
9. The outboard motor according to claim 6, wherein the sub-cylinder includes a single sub-cylinder, and both of the pair of oil chambers of the hydraulic cylinder are in communication with the sub-oil chamber of the single sub-cylinder.
10. The outboard motor according to claim 6, wherein the sub-cylinder includes only a single sub-cylinder; the gas chamber includes a pair of gas chambers; the sub-oil chamber includes a pair of sub-oil chambers; the pair of oil chambers of the hydraulic cylinder are in communication with the sub-oil chambers corresponding to the pair of oil chambers, respectively; and the free piston includes a rod piston that includes a gas piston and a pair of rods, the gas piston divides the pair of gas chambers from each other, the pair of rods extend from the gas piston to both sides in a sliding direction of the free piston, and each of the pair of rods functions as a hydraulic piston defining a portion of the sub-oil chamber corresponding to the rod.
11. The outboard motor according to claim 6, further comprising an on-off valve to interrupt a flow of hydraulic oil between at least one of the pair of oil chambers of the hydraulic cylinder and the sub-oil chamber of the sub-cylinder.
12. The outboard motor according to claim 11, further comprising: a prime mover to rotate a propeller; a detector to detect a rotation speed of the prime mover; and a controller configured or programmed to shut off the on-off valve when a rotation speed detected by the detector exceeds a predetermined threshold value.
13. An outboard motor comprising: an outboard motor main body; a mount assembly to mount the outboard motor main body to a hull, the mount assembly including a clamp bracket to be fixed to the hull and a swivel bracket that is coupled to the clamp bracket through a tilt shaft, turnable around the tilt shaft, and supports the outboard motor main body; a hydraulic cylinder between the clamp bracket and the swivel bracket to turn the swivel bracket around the tilt shaft with respect to the clamp bracket; and a gas damper to damp an external force that acts in a telescopic direction of the hydraulic cylinder; wherein the hydraulic cylinder includes a cylinder main body, a piston slidable in the cylinder main body and defining a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers, and is operable to extend and contract the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers; the piston includes a hollow piston including a cylindrical inner peripheral surface that defines a gas chamber filled with gas; and the gas damper includes a double piston structure including the hollow piston and an internal piston housed in the hollow piston slidable on the inner peripheral surface, that is coupled to the rod, and partitions the gas chamber into a pair of gas chambers.
14. The outboard motor according to claim 13, wherein the hollow piston includes a contact portion contactable with the internal piston to regulate a minimum volume of the gas chamber by regulating a position of a stroke end of the internal piston with respect to the hollow piston.
15. The outboard motor according to claim 1, wherein the hydraulic cylinder includes a cylinder main body, a piston slidable in the cylinder main body and defining a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers, and is operable to extend and contract the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers; and the gas damper includes a pair of independent foam structures located in the pair of oil chambers, respectively.
16. The outboard motor according to claim 1, wherein the hydraulic cylinder includes a cylinder main body, a piston slidable in the cylinder main body and defining a pair of oil chambers in the cylinder main body, and a rod that extends outwardly from the cylinder main body through one of the pair of oil chambers, and is operable to extend and contract the rod by selectively supplying hydraulic oil from a hydraulic circuit to either one of the pair of oil chambers; and the gas damper

includes a movable partition that defines a gas chamber filled with gas in the hydraulic cylinder or in a sub-cylinder that is hydraulically connected to the hydraulic cylinder.
